Clinical Study Summary Report (CSSR) Document Status: Draft | **Version:** 1.0 | **Date:** 07-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Study Evaluating the Efficacy, Safety, Pharmacokinetics and Pharmacodynamics of Crovalimab in Participants With Paroxysmal Nocturnal Hemoglobinuria (PNH) Not Previously Treated With Complement Inhibition **Official Title:** A Phase III, Multicenter, Single Arm Study Evaluating the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Crovalimab in Patients With Paroxysmal Nocturnal Hemoglobinuria (PNH) Not Previously Treated With Complement Inhibition **Short Title/ Acronym:** COMMODORE 3 **Protocol Number:** YO42311 **NCT Number:** NCT04654468 **Sponsor/Company Name:** Hoffmann-La Roche / Roche **Investigational Product:** Crovalimab (Novel C5 Inhibitor) **Phase of Development:** Phase III **Medical Monitor:** Hoffmann-La Roche Clinical Trial Medical Monitor

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy of crovalimab in achieving hemolysis control and transfusion avoidance in patients with PNH who have not been previously treated with complement inhibitor therapy. **Secondary Objectives:** To evaluate the safety, pharmacokinetics, pharmacodynamics, the percentage of participants with breakthrough hemolysis, stabilized hemoglobin, and improvements in fatigue (using the FACIT-Fatigue scale) of crovalimab therapy.

2.2 Background & Rationale To address the unmet medical need for complement C5 inhibitor-naive patients with Paroxysmal Nocturnal Hemoglobinuria (PNH). PNH is a lifelong, life-threatening disease characterized by complement-mediated hemolysis. Crovalimab is a novel anti-C5 recycling monoclonal antibody engineered for low-volume, subcutaneous every-4-week administration, allowing for self-administration. This study evaluates crovalimab to provide an efficacious and well-tolerated treatment option that significantly reduces the treatment burden for this patient population.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING)

3.2 Planned Enrollment & Duration Targeted Enrollment (N): 51 enrolled **Number of Sites:** 5 to 7 hospital sites in China

Estimated Study Start: 2021 **Estimated Primary Completion:** 2022 (Primary analysis clinical cutoff: February 2022)

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 12 years and body weight ≥ 40 kg at screening. 2. Willingness and ability to comply with all study visits and procedures. 3. Documented diagnosis of PNH confirmed by high-sensitivity flow cytometry, with Lactate Dehydrogenase (LDH) levels $\ge 2x$ the Upper Limit of Normal (ULN) at screening. 4. Participants who have at least four packed red blood cell (pRBC) transfusions during the 12 months prior to screening (documented in the medical record). 5. Presence of one or more of the PNH-related signs or symptoms within 3 months of screening. 6. Vaccination against *Neisseria meningitidis* serotypes A, C, W, and Y < 3 years prior to initiation of study treatment (Day 1). 7. Vaccination against *Haemophilus influenzae* type B and *Streptococcus pneumoniae* according to national vaccination recommendations. 8. For participants receiving other therapies (e.g., immunosuppressants, corticosteroids): stable dose for ≥ 28 days prior to screening and up to the first drug administration. 9. Adequate hepatic and renal function. 10. Women of childbearing potential: agreement to remain abstinent (refrain from heterosexual intercourse) or use contraception during the treatment period and for 46 weeks (approximately 10.5 months) after the final dose of crovalimab. 11. Platelet count $\ge 30,000$ per cubic millimeter (mm^3) at screening. 12. Absolute Neutrophil Count (ANC) $> 500/\mu L$ at screening.

4.2 Exclusion Criteria 1. Current or previous treatment with a complement inhibitor. 2. History of allogeneic bone marrow transplantation. 3. History of *Neisseria meningitidis* infection within 6 months prior to screening and up to first drug administration. 4. Known or suspected immune or hereditary complement deficiency. 5. Known HIV infection with cluster of differentiation 4 (CD4) count < 200 cells/μL within 24 weeks prior to screening. 6. Infection requiring hospitalization or treatment with IV antibiotics within 28 days prior to screening and up to the first drug administration, or oral antibiotics within 14 days prior to screening and up to the first drug administration. 7. Active systemic bacterial, viral, or fungal infection within 14 days before first drug administration. 8. Presence of fever ($\ge 38^\circ C$) within 7 days before the first drug administration. 9. Splenectomy < 6 months before screening. 10. History of malignancy within 5 years prior to screening and up to the first drug administration. 11. Pregnant or intending to become pregnant during the study or within 46 weeks (10.5 months) after the final dose of study treatment. 12. Participation in another interventional treatment study with an investigational agent or use of any experimental therapy within 28 days of screening or within 5 half-lives of that investigational product, whichever is greater.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Weight-based tiered-dosing schedule consisting of loading doses followed by maintenance doses. **Route of Administration:** One intravenous (IV) loading dose, followed by four subcutaneous (SC) loading doses, and subsequent subcutaneous maintenance doses. **Duration of Treatment:** Evaluated over a 24-week period for primary analysis (followed by an extended treatment period).

5.2 Concomitant & Prohibited Therapy Permitted: Stable doses of concomitant therapies (e.g., immunosuppressants, corticosteroids) if maintained for ≥ 28 days prior to screening and up to the first drug administration. **Prohibited:** Prior complement inhibitor therapy (patients must be complement inhibitor-naive).

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening** | Day -28 to -1 | Informed Consent, Medical & Transfusion History, Flow Cytometry, LDH Labs, Vaccination Verification | | **Baseline** | Day 1 | First IV Loading Dose, PK/PD, Safety Labs, Baseline FACIT-Fatigue | | **Interim** | Weeks 1 to 24 | Subcutaneous loading and Q4W maintenance dosing, LDH monitoring, Transfusion tracking, Efficacy/Safety Assessments | | **End of Study** | Week 25 | Final primary endpoint evaluation (LDH hemolysis control and Transfusion avoidance check) |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoints 1. **Mean Percentage of Participants With Hemolysis Control:** Estimated using a Generalized Estimating Equation (GEE) for population-average percentage of participants with hemolysis control (LDH $\le 1.5x$ ULN) from Week 5 through Week 25, taking into account intra-patient and inter-patient correlation. 2. **Difference in Percentage of Participants With Transfusion Avoidance (TA) From Baseline Through Week 25 and Within 24 Weeks Prior to Screening:** TA defined as participants who were pRBC transfusion-free and did not require transfusion per protocol-specified guidelines.

7.2 Secondary & Exploratory Endpoints - Percentage of participants with Breakthrough Hemolysis (BTH). - Percentage of participants with Stabilized Hemoglobin. - Change from baseline in fatigue in adults aged ≥ 18 years (e.g., FACIT-Fatigue scale score). - Pharmacokinetics (PK) and Pharmacodynamics (PD) profiling of crovalimab. - Incidence of treatment-emergent Adverse Events (AEs).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard clinical monitoring including tracking of patient fatigue, infections, administration site reactions, and regular blood work. **Serious Adverse Events (SAEs):** Evaluated strictly (e.g., specific tracking for meningococcal infections or treatment-related severe outcomes). **Data Monitoring Committee (DMC):** Supervised per sponsor guidelines to evaluate safety signals throughout the trial duration.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Efficacy evaluations applied to patients receiving ≥ 1 crovalimab dose and having ≥ 1 central LDH assessment post-dose (ITT/Efficacy set). Safety set for all patients receiving treatment. **Sample Size Justification:** A sample of 51 enrolled patients adequately powers the trial to demonstrate statistical significance against historical baseline rates for transfusion avoidance. **Handling of Missing Data:** Regulated via sponsor-defined statistical imputation procedures. 95% Confidence Interval (CI) for TA differences calculated using the Newcombe method.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Scheduled onsite and remote clinical monitoring to ensure protocol adherence, laboratory integrity, and accurate reporting of severe adverse events. **Audit Trail:** Robust tracking of eCRF entries, central laboratory data management (e.g., LDH results), and data clarification procedures.

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11. Study Identifiers & Governance

Primary ID: YO42311 **Posting ID:** 669863048493186011 **Registry Number:** NCT04654468 **Sponsor/Lead Organization:** Roche / Hoffmann-La Roche **Study Title:** COMMODORE 3 **Official Regulatory Title:** A Phase III, Multicenter, Single Arm Study Evaluating the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Crovalimab in Patients With Paroxysmal Nocturnal Hemoglobinuria (PNH) Not Previously Treated With Complement Inhibition

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12. Clinical Framework (PNH Specific)

Primary Condition: Paroxysmal Nocturnal Hemoglobinuria (PNH) **Disease Areas:** Paroxysmal Nocturnal Hemoglobinuria **Diagnosis**

Threshold: PNH Clone via flow cytometry and LDH $\ge 2x$ ULN **Medical Methodology:** Complement C5 Inhibition **Optimization Target:** Sustained Hemolysis Control (LDH $\le 1.5x$ ULN) and Transfusion Avoidance

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13. Study Procedures & Methodology

Management Pathway: Subcutaneous maintenance treatment for complement inhibition **Education Protocol (HCP):** Crovalimab SC dosing schedule and weight-tiered administration **Education Protocol (Patient):** Education on vaccination requirements and potential SC self-administration **Quality Audit Mechanism:** Routine LDH testing, AE monitoring, and strict transfusion logging

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14. Observation & Follow-Up Schedule

Total Duration: 25 Follow-up weeks for Primary Evaluation **Onsite Visit Cadence:** Weeks 1, 2, 3, 5, 9, 13, 17, 21, and 25 (Adjusted for loading & maintenance phases) **Remote Monitoring Frequency:** Every 4 weeks (Q4W) concurrent with SC maintenance dosing **Checklist Review Interval:** Every 4 to 12 weeks for FACIT-Fatigue and Safety Assessments

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	**[DD-MMM-YYYY]**
Clinical Operations Lead	**[Name]**	___	***[DD-MMM-YYYY]***
Lead Statistician	***[Name]***	_____	[DD-MMM-YYYY]